

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275445

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

KIM; SEONG-IL et al.

DISPLAY DEVICE

Abstract

A display device can include a display panel, and a switchable reflective layer having an electrode layer and an electroactive material pattern and positioned at a side of the display panel. In a first mode in the electrode layer, the electroactive material pattern has a first thickness, a first width and a first pitch. In a second mode in the electrode layer, the electroactive material pattern has a second thickness, a second width and a second pitch. The electroactive material pattern satisfies at least one of i) the second thickness being different from the first thickness, ii) the second width being different from the first width, and iii) the second pitch being different from the first pitch.

Inventors: KIM; SEONG-IL (Paju-si, KR), JEON; HYUN-WOO (Paju-si, KR), OH; YOUNG-MIN (Paju-si, KR)

Applicant: LG Display Co., Ltd. (Seoul, KR)

Family ID: 1000008281912

Assignee: LG Display Co., Ltd. (Seoul, KR)

Appl. No.: 18/946552

Filed: November 13, 2024

Foreign Application Priority Data

KR 10-2024-0025749

Feb. 22, 2024

Publication Classification

Int. Cl.: H10K59/80 (20230101); C09K19/38 (20060101); G02F1/1335 (20060101); G02F1/19 (20190101); H10K59/50 (20230101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Korean Patent Application No. 10-2024-0025749, filed in the Republic of Korea on Feb. 22, 2024, the entire contents of which is hereby expressly incorporated by reference into the present application.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a display device, and more particularly, to a display device being capable of controlling an ambient light reflection.

Discussion of the Related Art

[0003] As information technology and mobile communication technology have been developed, a display device being capable of displaying a visual image has also been developed. Flat panel display devices, such as a liquid crystal display (LCD) device, a plasma display panel device (PDP) and an organic light emitting display (OLED) device, are developed and used.

[0004] In general, an LCD device includes a liquid crystal panel, which includes upper and lower substrates and a liquid crystal layer therebetween. The liquid crystal layer is driven by an electric field induced between a pixel electrode and a common electrode to display images.

[0005] An OLED device includes an organic emitting diode which includes an anode, a cathode and an organic emitting layer therebetween. In the organic emitting diode, holes and electrons respectively from the anode and the cathode are combined such that light is emitted from the organic emitting layer to display images.

[0006] Recently, a display device is widely used in vehicles, but the internal design of the vehicle and the external design of the display device may not be harmonized with each other. Accordingly, there can be a problem of visual quality deterioration.

SUMMARY OF THE DISCLOSURE

[0007] Accordingly, embodiments of the present disclosure are directed to a display device that substantially obviates one or more of the problems associated with the limitations and disadvantages of the related art.

[0008] An object of the present disclosure is to provide a display device being capable of controlling an ambient light reflection to improve harmony between the internal design of the vehicle and the external design of the display device.

[0009] Additional features and aspects will be set forth in the description that follows, and in part will be apparent from the description, or can be learned by practice of the present disclosure concepts provided herein. Other features and aspects of the present disclosure concepts can be realized and attained by the structure particularly pointed out in the written description, or derivable therefrom, and the claims hereof as well as the appended drawings.

[0010] To achieve these and other advantages in accordance with the purpose of the embodiments of the present disclosure, as described herein, an aspect of the present disclosure is a display device comprising a display panel; and a switchable reflective layer including an electrode layer and an electroactive material pattern and positioned at a side of the display panel, wherein in a first mode in the electrode layer, the electroactive material pattern has a first thickness, a first width and a first pitch, wherein in a second mode in the electrode layer, the electroactive material pattern has a second thickness, a second width and a second pitch, and wherein the electroactive material pattern satisfies at least one of i) the second thickness being different from the first thickness, ii) the second

width being different from the first width, and iii) the second pitch being different from the first pitch.

[0011] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the inventive concepts as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The accompanying drawings, which are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this application, illustrate embodiments of the present disclosure and together with the description serve to explain principles of the present disclosure.

[0013] FIG. 1 is a schematic cross-sectional view of a display device according to a first embodiment of the present disclosure.

[0014] FIGS. 2A and 2B are pictures showing the display device according to the first embodiment with on and off states, respectively.

[0015] FIG. 3 is schematic cross-sectional view of a display device according to a second embodiment of the present disclosure.

[0016] FIG. 4A is a schematic circuit diagram of an organic light emitting diode panel according to aspects of the present disclosure.

[0017] FIG. 4B is a schematic cross-sectional view of an organic light emitting diode panel according to aspects of the present disclosure.

[0018] FIG. 5 is a schematic cross-sectional view of a liquid crystal panel according to aspects of the present disclosure.

[0019] FIGS. 6A and 6B are schematic plane views of an electrode layer of a switchable reflective layer, respectively.

[0020] FIGS. 7A and 7B are schematic cross-sectional views illustrating a shape change of an electroactive material of a switchable reflective layer according to an applied voltage.

[0021] FIG. 8 is a schematic plane view showing an electroactive material pattern of a switchable reflective layer.

[0022] FIGS. 9A and 9B are graphs showing reflection and transmittance of a visible ray in a switchable reflective layer.

[0023] FIG. 10 is a schematic plane view showing an electroactive material pattern of a switchable reflective layer.

[0024] FIGS. 11A and 11B are graphs showing reflection and transmittance of a visible ray in a switchable reflective layer.

[0025] FIGS. 12A and 12B are graphs showing reflection and transmittance of a visible ray in a switchable reflective layer.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0026] Reference will now be made in detail to aspects of the present disclosure, examples of which can be illustrated in the accompanying drawings. In the following description, when a detailed description of well-known functions or configurations related to this document is determined to unnecessarily cloud a gist of the inventive concept, the detailed description thereof will be omitted. The progression of processing steps and/or operations described is an example; however, the sequence of steps and/or operations is not limited to that set forth herein and can be changed as is known in the art, with the exception of steps and/or operations necessarily occurring in a particular order. Like reference numerals designate like elements throughout. Names of the respective elements used in the following explanations are selected only for convenience of writing

the disclosure and can be thus different from those used in actual products.

[0027] Advantages and features of the present disclosure and methods of achieving them will be apparent with reference to the aspects described below in detail with the accompanying drawings. However, the present disclosure is not limited to the aspects disclosed below, but can be realized in a variety of different forms, and only these aspects allow the disclosure of the present disclosure to be complete. The present disclosure is provided to fully inform the scope of the disclosure to the skilled in the art of the present disclosure.

[0028] The shapes, sizes, proportions, angles, numbers, and the like disclosed in the drawings for explaining the aspects of the present disclosure are illustrative, and the present disclosure is not limited to the illustrated matters. The same reference numerals refer to the same elements throughout the disclosure. In addition, in describing the present disclosure, if it is determined that a detailed description of the related known technology unnecessarily obscure the subject matter of the present disclosure, the detailed description thereof can be omitted. When ‘including’, ‘having’, ‘consisting’, and the like are used in this disclosure, other parts can be added unless ‘only’ is used. When a component is expressed in the singular, cases including the plural are included unless specific statement is described.

[0029] In construing an element, the element is construed as including an error or tolerance range although there is no explicit description of such an error or tolerance range.

[0030] In describing a position relationship, for example, when a position relation between two parts is described as, for example, “on,” “over,” “under,” and “next,” one or more other parts can be disposed between the two parts unless a more limiting term, such as “just” or “direct(ly)” is used.

[0031] In describing a time relationship, for example, when the temporal order is described as, for example, “after,” “subsequent,” “next,” and “before,” a case that is not continuous can be included unless a more limiting term, such as “just,” “immediate(ly),” or “direct(ly)” is used.

[0032] It will be understood that, although the terms “first,” “second,” etc. can be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure.

[0033] Features of various aspects of the present disclosure can be partially or overall coupled to or combined with each other, and can be variously inter-operated with each other and driven technically as those skilled in the art can sufficiently understand. The aspects of the present disclosure can be carried out independently from each other, or can be carried out together in co-dependent relationship. Further, the term “can” fully encompasses all the meanings and coverages of the term “may.”

[0034] Reference will now be made in detail to some of the examples and embodiments, which are illustrated in the accompanying drawings. All the components of each display device according to all embodiments of the present disclosure are operatively coupled and configured.

[0035] FIG. 1 is a schematic cross-sectional view of a display device according to a first embodiment of the present disclosure.

[0036] As shown in FIG. 1, a display device **100** of the present disclosure includes a display panel **110** and a colored film **120**.

[0037] For example, the display panel **110** can be an organic light emitting diode panel or a liquid crystal panel.

[0038] The colored film **120** is disposed over a display surface of the display panel **110**. For example, the colored film **120** has a specific color to form the external appearance of the display device **100**.

[0039] Referring to FIG. 2A, which is a picture showing the display device according to the first embodiment with on state, the display device **100** has a specific color by the colored film **120** has a specific color so that the display device **100** and the design of the surrounding space can be

harmonized with each other.

[0040] However, Referring to FIG. 2B, which is a picture showing the display device according to the first embodiment with off state, there are problems of low luminance and image distortion in the display device **100** by the colored film **120**.

[0041] FIG. 3 is schematic cross-sectional view of a display device according to a second embodiment of the present disclosure.

[0042] As shown in FIG. 3, a display device **200** includes a display panel **210** and a switchable reflective layer **220** including an electrode layer **240** and an electroactive material pattern **260** and positioned at a side of the display panel **210**.

[0043] The display panel **210** is disposed under the electrode layer **240** and displays an image.

[0044] In an aspect of the present disclosure, the display panel **210** can be an organic light emitting diode panel.

[0045] Referring to FIG. 4A, which is a schematic circuit diagram of an organic light emitting diode panel of the present disclosure, in the organic light emitting diode panel, a gate line GL and a data line DL, which can cross each other to define a pixel region P, and a power line PL can be formed in an organic light display device. A switching thin film transistor (TFT) Ts, a driving thin film transistor (TFT) Td, a storage capacitor Cst, and an OLED D can be formed in the pixel region P. The pixel region P can include a red pixel region, a green pixel region, and a blue pixel region.

[0046] The switching thin film transistor Ts can be connected to the gate line GL and the data line DL, and the driving thin film transistor Td and the storage capacitor Cst can be connected between the switching thin film transistor Ts and the power line PL. The OLED D can be connected to the driving thin film transistor Td. When the switching thin film transistor Ts is turned on by the gate signal applied through the gate line GL, the data signal applied through the data line DL can be applied to a gate electrode of the driving thin film transistor Td and one electrode of the storage capacitor Cst through the switching thin film transistor Ts.

[0047] The driving thin film transistor Td can be turned on by the data signal applied to the gate electrode so that a current proportional to the data signal can be supplied from the power line PL to the OLED D through the driving thin film transistor Td. The OLED D can emit light having a luminance proportional to the current flowing through the driving thin film transistor Td. In this case, the storage capacitor Cst can be charged with a voltage proportional to the data signal so that the voltage of the gate electrode in the driving thin film transistor Td can be kept constant or similar during one frame. Therefore, the organic light emitting display device can display a desired image.

[0048] Referring to FIG. 4B, which is a schematic cross-sectional view of an organic light emitting diode panel of the present disclosure, the organic light emitting diode panel **210** can include a substrate **312**, a TFT Tr over the substrate **312**, a planarization layer **350** covering the TFT Tr and an OLED D disposed on the planarization layer **350** and connected to the TFT Tr. For example, the organic light emitting diode panel **210** can include a red pixel region, a green pixel region, and a blue pixel region.

[0049] The substrate **312** can be a glass substrate or a flexible substrate. For example, the flexible substrate can be a polyimide (PI) substrate, a polyethersulfone (PES) substrate, a polyethylenenaphthalate (PEN) substrate, a polyethylene terephthalate (PET) substrate or a polycarbonate (PC) substrate.

[0050] A buffer layer **320** can be formed on the substrate, and the TFT Tr can be formed on the buffer layer **320**. The buffer layer **320** can be omitted. The buffer layer **320** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride.

[0051] A semiconductor layer **322** can be formed on the buffer layer **320**. The semiconductor layer **322** can include an oxide semiconductor material or polycrystalline silicon.

[0052] When the semiconductor layer **322** includes the oxide semiconductor material, a light-shielding pattern can be formed under the semiconductor layer **322**. The light to the semiconductor

layer **322** can be shielded or blocked by the light-shielding pattern such that thermal degradation of the semiconductor layer **322** can be prevented or reduced. On the other hand, when the semiconductor layer **322** includes polycrystalline silicon, impurities can be doped into both sides of the semiconductor layer **322**.

[0053] A gate insulating layer **324** can be formed on the semiconductor layer **322**. The gate insulating layer **324** can be formed of an inorganic insulating material such as silicon oxide or silicon nitride.

[0054] A gate electrode **330**, which can be formed of a conductive material, e.g., metal, can be formed on the gate insulating layer **324** to correspond to a center of the semiconductor layer **322**.

[0055] In FIG. **4B**, the gate insulating layer **324** can be formed on an entire surface of the substrate **312**. Alternatively, the gate insulating layer **324** can be patterned to have the same shape as the gate electrode **330**.

[0056] An interlayer insulating layer **332**, which can be formed of an insulating material, can be formed on the gate electrode **330**. The interlayer insulating layer **332** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

[0057] The interlayer insulating layer **332** can include first and second contact holes **334** and **336** exposing both sides of the semiconductor layer **322**. The first and second contact holes **334** and **336** may not cover a portion of the surface of the semiconductor layer **322** that is nearer to the opposing ends than to a center of the semiconductor layer **322**. The first and second contact holes **334** and **336** can be positioned at both sides of the gate electrode **330** to be spaced apart from the gate electrode **330**. In FIG. **4B**, the first and second contact holes **334** and **336** can be formed through the interlayer insulating layer **332** and the gate insulating layer **324**. Alternatively, when the gate insulating layer **324** is patterned to have the same shape as the gate electrode **330**, the first and second contact holes **334** and **336** can be formed only through the interlayer insulating layer **332**.

[0058] A source electrode **340** and a drain electrode **342**, which can be formed of a conductive material, e.g., metal, can be formed on the interlayer insulating layer **332**. The source electrode **340** and the drain electrode **342** can be spaced apart from each other with respect to the gate electrode **330** and can contact both sides of the semiconductor layer **322** through the first and second contact holes **334** and **336**, respectively.

[0059] The semiconductor layer **322**, the gate electrode **330**, the source electrode **340**, and the drain electrode **342** can constitute the TFT Tr. The TFT Tr can serve as a driving element. For example, the TFT Tr can correspond to the driving TFT Td (of FIG. **4A**).

[0060] In the TFT Tr, the gate electrode **330**, the source electrode **340**, and the drain electrode **342** can be positioned on the semiconductor layer **322**. For example, the TFT Tr can have a coplanar structure.

[0061] Alternatively, in the TFT Tr, the gate electrode can be positioned under the semiconductor layer, and the source and drain electrodes can be positioned on the semiconductor layer such that the TFT Tr can have an inverted staggered structure. In this instance, the semiconductor layer can include amorphous silicon.

[0062] The gate line and the data line can cross each other to define the pixel region, and the switching TFT can be formed to be connected to the gate and data lines. The switching TFT can be connected to the TFT Tr as the driving element. In addition, the power line, which can be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame can be further formed.

[0063] A planarization layer (or a passivation layer) **350** can be formed to cover the TFT Tr. The planarization layer **350** can provide a flat top surface and includes a drain contact hole **352** exposing the drain electrode **342** of the TFT Tr. The planarization layer **350** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

[0064] The OLED D is disposed on the planarization layer **350** and includes a first electrode **360**, which is connected to the drain electrode **342** of the TFT Tr, an organic light emitting layer **362** on the first electrode **360** and a second electrode **364** on the organic light emitting layer **362**. The OLED D is positioned in each of the red, green and blue pixel regions and emits red, green and blue light at the red, green and blue pixel regions, respectively.

[0065] The first electrode **360** can be separately formed in each pixel region. The first electrode **360** can be an anode and can include a transparent conductive oxide material layer formed of a conductive material, e.g., a transparent conductive oxide (TCO), having a relatively high work function. For example, the transparent conductive oxide material layer of the first electrode **360** can include at least one of indium-tin-oxide (ITO) indium-zinc-oxide (IZO), indium-tin-zinc oxide; ITZO), tin oxide (SnO), zinc oxide (ZnO), indium-copper-oxide (ICO) and Al:ZnO (AZO).

[0066] The first electrode **360** can have a single-layered structure of the transparent conductive oxide material layer. Namely, the first electrode **360** can be a transparent electrode.

[0067] Alternatively, the first electrode **360** can further include a reflection layer to have a double-layered structure or a triple-layered structure. Namely, the first electrode **360** can be a reflection electrode.

[0068] For example, the reflection layer can be formed of silver (Ag), an alloy of Ag with at least one of palladium (Pd), copper (Cu), indium (In) and neodymium (Nd) or aluminum-palladium-copper (APC) alloy. In an aspect of the present disclosure, the first electrode **360** can have a double-layered structure of Ag/ITO or APC/ITO or a triple-layered structure of ITO/Ag/ITO or ITO/APC/ITO. However, embodiments of the present disclosure are not limited to such examples.

[0069] A bank layer **366** can be formed on the planarization layer **350** to cover an edge of the first electrode **360**. For example, the bank layer **366** can be positioned at a boundary of the pixel region and exposes a center of the first electrode **360** in the pixel region.

[0070] The organic light emitting layer **362** including an emitting material layer (EML) is formed on the first electrode **360**. The organic light emitting layer **362** can have a multi-layered structure including at least one of a hole injection layer (HIL), a hole transporting layer (HTL), an electron blocking layer (EBL), a hole blocking layer (HBL), an electron transporting layer (ETL), and an electron injection layer (EIL).

[0071] In an aspect of the present disclosure, the organic light emitting layer **362** can emit red, green and blue light at the red, green and blue pixel regions, respectively. Alternatively, the organic light emitting layer **362** can emit white light at the red, green and blue pixel regions.

[0072] A second electrode **364** is formed over the substrate **312** where the organic light emitting layer **362** is formed. The second electrode **364** can cover an entire surface of the display area and can be formed of a conductive material having a relatively low work function to serve as a cathode. For example, the second electrode **364** can be formed of aluminum (Al), magnesium (Mg), calcium (Ca), silver (Ag) or their alloy, e.g., MgAg, or a combination thereof.

[0073] In the top-emission type organic light emitting diode panel **210**, the first electrode **360** is a reflection electrode, and the second electrode **364** has a thin profile (small thickness) to provide a light transmittance property (or a semi-transmittance property). In the bottom-emission type organic light emitting diode panel **210**, the first electrode **360** is a transparent electrode, and the second electrode **364** is a reflection electrode.

[0074] In the top-emission type organic light emitting diode panel **210**, a capping layer can be further formed on the second electrode **364**. The optical efficiency of the organic light emitting diode panel **210** can be further improved by the capping layer.

[0075] An encapsulation layer (e.g., an encapsulation film) **370** can be formed on the second electrode **364** to prevent penetration of moisture into the OLED D. The encapsulation layer **370** can include a first inorganic insulating layer **372**, an organic insulating layer **374**, and a second inorganic insulating layer **376** sequentially stacked. However, embodiments of the present disclosure are not limited to such examples. The encapsulation layer **370** can be omitted.

[0076] In the bottom-emission type organic light emitting diode panel **210**, a metal encapsulation plate can be further disposed on the second electrode **364** or the encapsulation layer **370**. For example, metal encapsulation plate can be attached onto the second electrode **364** or the encapsulation layer **370** using an adhesive layer.

[0077] The organic light emitting diode panel **210** can further include a color filter layer corresponding to the red, green and blue pixel regions. In the top-emission type organic light emitting diode panel **210**, the color filter layer can be positioned over the OLED D, e.g., over the second electrode **364** or the encapsulation layer **370**. In the bottom-emission type organic light emitting diode panel **210**, the color filter layer can be positioned between the OLED D and the substrate **312**, e.g., between the interlayer insulating layer **332** and the planarization layer **350**.

[0078] The color filter layer can include red, green and blue color filter patterns respectively corresponding to the red, green and blue pixel regions. The red color filter pattern can include at least one of a red dye and a red pigment, the green color filter pattern can include at least one of a green dye and a green pigment, and the blue color filter pattern can include at least one of a blue dye and a blue pigment.

[0079] The organic light emitting diode panel **210** can further include a polarization plate for reducing an ambient light reflection. For example, the polarization plate can be a circular polarization plate. In the bottom-emission type organic light emitting diode panel **210**, the polarization plate can be disposed under the substrate **312**. In the top-emission type organic light emitting diode panel **210**, the polarization plate can be disposed on or over the encapsulation layer **370**.

[0080] In an aspect of the present disclosure, the display panel **210** can be a liquid crystal panel.

[0081] Referring to FIG. 5, which is a schematic cross-sectional view of a liquid crystal panel of the present disclosure, the liquid crystal panel **210** includes first and second substrates **412** and **450**, which face each other, and a liquid crystal layer **460**, which includes liquid crystal molecules **462**, therebetween.

[0082] Each of the first and second substrates **412** and **450** can be a glass substrate or a flexible substrate. For example, the flexible substrate can be one of polyimide (PI) substrate, polyethersulfone (PES) substrate, polyethylenenaphthalate (PEN) substrate, polyethylene terephthalate (PET) substrate and polycarbonate (PC) substrate.

[0083] A first buffer layer **420** is formed on the first substrate **412**, and a TFT Tr is formed on the first substrate **412**. The first buffer layer **420** can be omitted. The first buffer layer **420** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride.

[0084] A gate electrode **422** is formed on the first buffer layer **420**, and a gate insulating layer **424** is formed on the gate electrode **422**. In addition, a gate line, which is connected to the gate electrode **422**, is formed on the first buffer layer **420**.

[0085] A semiconductor layer **426**, which corresponds to the gate electrode **422**, is formed on the gate insulating layer **424**. The semiconductor layer **426** can include an oxide semiconductor material. Alternatively, the semiconductor layer can include an active layer of intrinsic amorphous silicon and an ohmic contact layer of impurity-doped amorphous silicon.

[0086] A source electrode **430** and a drain electrode **432**, which are spaced apart from each other, are formed on the semiconductor layer **426**. In addition, a data line, which is electrically connected to the source electrode **430** and crosses the gate line to define a pixel region, is formed.

[0087] The gate electrode **422**, the semiconductor layer **426**, the source electrode **430** and the drain electrode **432** constitute the TFT Tr.

[0088] A planarization layer **434**, which includes a drain contact hole **436** exposing the drain electrode **432**, is formed on the TFT Tr. The planarization layer **434** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

[0089] A pixel electrode **440**, which is connected to the drain electrode **432** through the drain

contact hole **436**, and a common electrode **442**, which is alternately arranged with the pixel electrode **440**, are formed on the planarization layer **434**.

[0090] Each of the pixel electrode **440** and the common electrode **442** can be formed of a transparent conductive material. For example, each of the pixel electrode **440** and the common electrode **442** can be formed of one of indium-tin-oxide (ITO) indium-zinc-oxide (IZO), indium-tin-zinc oxide; ITZO), tin oxide (SnO), zinc oxide (ZnO), indium-copper-oxide (ICO) and Al:ZnO (AZO).

[0091] A second buffer layer **452** is formed on the second substrate **450**. The second buffer layer **452** can be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride.

[0092] A black matrix **454**, which shields a non-display region such as the TFT Tr, the gate line and the data line, is formed on the second buffer layer **452**. In addition, a color filter layer **456**, which corresponds to the pixel region, is formed on the second buffer layer **452**. The second buffer layer **452** and the black matrix **454** can be omitted.

[0093] The display panel **210** can include red, green and blue pixel regions, and the color filter layer **456** can include red, green and blue color filter patterns respectively corresponding to the red, green and blue pixel regions. The red color filter pattern can include at least one of red dye and red pigment. The green color filter pattern can include at least one of green dye and green pigment. The blue color filter pattern can include at least one of blue dye and blue pigment.

[0094] The first and second substrates **412** and **450** are attached with the liquid crystal layer **460** therebetween. The liquid crystal molecules **462** of the liquid crystal layer **460** is driven by an electric field between the pixel and common electrode **440** and **442**.

[0095] In FIG. 5, the pixel electrode **440** and the common electrode **442** are disposed on the same layer, i.e., on the planarization layer **434**. Alternatively, the pixel electrode **440** and the common electrode **442** can be disposed at different layers. For example, the common electrode **442** can be disposed on the second electrode **450**.

[0096] First and second alignment layers can be formed over the first and second substrates **412** and **450** to be adjacent to the liquid crystal layer **460**. In addition, first and second polarization plates, which have perpendicular transmission axes, can be attached to an outer side of each of the first and second substrates **412** and **450**. Moreover, the liquid crystal panel **210** can further include a backlight unit under the first substrate **412**. For example, the backlight unit can include a light guide plate under the first substrate **412**, a light source at a side of the light guide plate, a reflection layer under the light guide plate and an optical sheet between the light guide plate and the first substrate **412**.

[0097] Referring to FIG. 3 again, the switchable reflective layer **220** is disposed at a display surface of the display panel **210**.

[0098] When the display panel **210** is the top-emission type organic light emitting diode panel, the switchable reflective layer **220** can be disposed on or over the encapsulation layer **370**.

Alternatively, when the display panel **210** is the bottom-emission type organic light emitting diode panel, the switchable reflective layer **220** can be disposed under the substrate **312**.

[0099] When the display panel **210** is the liquid crystal panel, the switchable reflective layer **220** can be disposed over the second substrate **450**.

[0100] In the switchable reflective layer **220**, the electrode layer **240** is positioned between the electroactive material pattern **260** and the display panel **210**.

[0101] The switchable reflective layer **220** can further include a base **230** between the display panel **210** and the electrode layer **240**. The base **230** provides a surface on which the electrode layer **240** is formed. The base **230** can be formed of polyethylene terephthalate (PET) or tri-acetyl cellulose (TAC).

[0102] The electrode layer **240** can be formed on the base **230**. Namely, the base **230** covers the display panel **210** and is disposed under the electrode layer **240**. The electrode layer **240** includes a plurality of first electrodes, which are spaced apart from each other, and a plurality of second

electrodes, which are alternately arranged with the plurality of the first electrodes. Alternatively, the base **230** can be omitted, and the electrode layer **240** can be formed directly on the display panel **210**.

[0103] Referring to FIG. **6A**, which is a schematic plane view of an electrode layer of a switchable reflective layer, the electrode layer **240** includes the plurality of first electrodes **242** and the plurality of second electrodes **244**.

[0104] The first electrodes **242** and the second electrodes **244** are alternately arranged with each other on the base **230** (of FIG. **3**) and along a first direction. Namely, the plurality of first electrodes **242** are arranged to be spaced apart from each other, and the plurality of second electrodes **244** are arranged between adjacent first electrodes **242** and to be spaced apart from the first electrode **242**.

[0105] Each of the first electrodes **242** and the second electrodes **244** can be include or formed of a transparent conductive material. For example, each of the first electrodes **242** and the second electrodes **244** can include at least one of indium-tin-oxide (ITO) indium-zinc-oxide (IZO), indium-tin-zinc oxide; ITZO), tin oxide (SnO), zinc oxide (ZnO), indium-copper-oxide (ICO) and Al:ZnO (AZO).

[0106] Referring to FIG. **6B**, which is a schematic plane view of an electrode layer of a switchable reflective layer, the electrode layer **240** can include the first electrodes **242**, which are spaced apart from each other in a first direction and a second direction, and the second electrodes **244**, which are spaced apart from each other in a first direction and a second direction and alternately arranged with the first electrodes **242**. Each of the first electrodes **242** and the second electrodes **244** can be include or formed of a transparent conductive material.

[0107] Referring to FIG. **3** again, an insulating layer **250** is formed on the electrode layer **240** including the first and second electrodes **242** and **244**. Namely, the insulating layer **250** covers the electrode layer **240** and is disposed under the electroactive material pattern **260**. The insulating layer **250** planarizes a step difference by the first and second electrodes **242** and **244** and can provide a flat top surface.

[0108] The insulating layer **250** can include or formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

[0109] The electroactive material pattern **260** is formed on the insulating layer **250**. In an aspect of the present disclosure, the insulating layer **250** can be omitted so that the electroactive material pattern **260** can be formed directly on the electrode layer **240**.

[0110] A cover window covering the electroactive material pattern **260** can be disposed on the electroactive material pattern **260**.

[0111] The electroactive material pattern **260** is substantially transparent, and at least one of a shape and a volume of the electroactive material pattern **260** is changed according to a voltage applied to the electrode layer **240**.

[0112] For example, a cation and/or an anion can be moved in the electroactive material pattern **260** according to a voltage applied to the electrode layer **240** so that a charge and/or a size of the electroactive material pattern **260** can be changed. As a result, a length and/or a shape of the electroactive material pattern **260** can be changed. Alternatively, the electroactive material pattern **260** can absorb an ion according to a voltage applied to the electrode layer **240** so that a volume of the electroactive material pattern **260** can be changed.

[0113] In an aspect of the present disclosure, a thickness, i.e., a height, of the electroactive material pattern **260** can be changed according to a voltage applied to the electrode layer **240**. In an aspect of the present disclosure, a width of the electroactive material pattern **260** can be changed according to a voltage applied to the electrode layer **240**. In other words, a distance, i.e., a pitch, between adjacent electroactive material patterns **260** can be changed.

[0114] The electroactive material pattern **260** can include an electroactive material. For example, the electroactive material pattern **260** can be an electroactive polymer.

[0115] In an aspect of the present disclosure, the electroactive material pattern **260** can include at least one of poly vinylidene fluoride (PVDF), poly vinylidene cyanide (PVDCN), poly vinylidene chloride (PVDCL), polypyrrole, polyacrylonitrile, polydimethylsiloxane (PDMS), amide polymer, urea polymer and carbonyl polymer.

[0116] For example, poly vinylidene fluoride (PVDF) can be represented by one of Formulas 1-1 to 1-3, and poly vinylidene cyanide (PVDCN) can be represented by Formula 2. The poly vinylidene chloride (PVDCL) can be represented by Formula 3, and the amide polymer can be represented by one of Formulas 4-1 and 4-2. The urea polymer can be represented by one of Formulas 5-1 and 5-2, and the carbonyl polymer can be represented by one of Formulas 6-1 and 6-2.

##STR00001##

[0117] In each of Formulas 1-1, 1-2 and 2 to 6-2, n is an integer of 100 to 20000. In Formula 1-3, a is a number of 0.7 to 0.8, and b is a number of 0.2 to 0.3. Here, it can be that $a+b=1$. As a result, the electroactive material can have a molecular weight of about 10000 to 100000.

[0118] The electroactive material can have a dipole moment. A degree of the shape change and or the size change of the electroactive material can depend on the in the intensity of the dipole moment.

[0119] Referring to FIG. 7A, which is a schematic cross-sectional view illustrating a shape change of an electroactive material of a switchable reflective layer according to an applied voltage, with FIGS. 6A and 6B, without a voltage to the electrode layer, the electroactive material pattern **260** has a first thickness t_1 , a first width w_1 and a first pitch p_1 , and with a voltage to the electrode layer, the electroactive material pattern **260** can have a second thickness t_2 smaller than the first thickness t_1 .

[0120] Namely, when a voltage is applied to the first and second electrodes **242** and **244** of the electrode layer **240**, a thickness of the electroactive material pattern **260** is decreased. Alternatively, when a voltage is applied to the first and second electrodes **242** and **244** of the electrode layer **240**, a thickness of the electroactive material pattern **260** can be increased.

[0121] Referring to FIG. 7B, which is a schematic cross-sectional view illustrating a shape change of an electroactive material of a switchable reflective layer according to an applied voltage, with FIGS. 6A and 6B, without a voltage to the electrode layer, the electroactive material pattern **260** has a first thickness t_1 , a first width w_1 and a first pitch p_1 , and with a voltage to the electrode layer, the electroactive material pattern **260** can have a second width w_2 greater than the first width w_1 and a second pitch p_2 smaller than the first pitch p_1 .

[0122] Namely, when a voltage is applied to the first and second electrodes **242** and **244** of the electrode layer **240**, a width of the electroactive material pattern **260** is increased. Alternatively, when a voltage is applied to the first and second electrodes **242** and **244** of the electrode layer **240**, a width of the electroactive material pattern **260** can be decreased so that the pitch of the electroactive material pattern **260** can be increased.

[0123] When the electroactive material pattern **260** has a first thickness t_1 , a first width w_1 and a first pitch p_1 , the ambient light is reflected by the electroactive material pattern **260**. Accordingly, the display device **200** can have a desired design. On the other hand, when the electroactive material pattern **260** has a second thickness t_2 , which is smaller than the first thickness t_1 , and/or a second pitch p_2 , which is smaller than the first pitch p_1 , the electroactive material pattern **260** does not reflect a visible light and has high transmittance. Namely, the transmittance and the reflectance of the switchable reflective layer **220** is changed according to a voltage to the electrode layer **240**.

[0124] In FIG. 7A, a thickness of the electroactive material pattern **260** is changed with constant width and pitch. In FIG. 7B, a width and a pitch of the electroactive material pattern **260** are changed with constant thickness. Alternatively, all of the thickness, the width and the pitch of the electroactive material pattern **260** can be changed according to a voltage to the electrode layer **240**.

[0125] In FIGS. 7A and 7B, without a voltage to the first and second electrodes **242** and **244** of the electrode layer **240**, the electroactive material pattern **260** has a first thickness t_1 , a first width w_1

and a first pitch p_1 so that a visible light is reflected by the electroactive material pattern **260**.
[0126] Alternatively, without a voltage to the first and second electrodes **242** and **244** of the electrode layer **240**, the visible light may not be reflected by the electroactive material pattern **260**, while with a voltage to the first and second electrodes **242** and **244** of the electrode layer **240**, at least one of the thickness, the width and the pitch of the electroactive material pattern **260** can be changed so that the electroactive material pattern **260** can have a first thickness t_1 , a first width w_1 and a first pitch p_1 . In this case, the electroactive material pattern **260** has high transmittance without a voltage to the first and second electrodes **242** and **244** of the electrode layer **240** so that the luminance of the display device **200** is increased. On the other hand, the electroactive material pattern **260** has high reflectance with a voltage to the first and second electrodes **242** and **244** of the electrode layer **240** so that the display device **200** has a desired design.

[0127] In the display device **200** of the present disclosure, the electroactive material pattern **260** with a first mode (e.g., without a voltage difference) in the electrode layer **240** and the electroactive material pattern **260** with a second mode (e.g., with a voltage difference) in the electrode layer **240** have a difference in at least one of a thickness, a width and a pitch. In other words, the electroactive material pattern **260** without a voltage difference in the electrode layer **240** has a first thickness, a first width and a first pitch, and the electroactive material pattern **260** with a voltage difference in the electrode layer **240** has a second thickness, a second width and a second pitch. The electroactive material pattern **260** satisfies at least one of the following conditions: i) the second thickness is different from the first thickness, ii) the second width is different from the first width, and iii) the second pitch is different from the first pitch. In the first mode, the same voltage, e.g., OV, is applied to the first and second electrodes **242** and **244** (of FIGS. 6A and 6B). In the second mode, the different voltages are applied to the first and second electrodes **242** and **244**.

[0128] For example, i) the second thickness is different from the first thickness and the second pitch is equal to the first pitch, and/or ii) the second thickness is equal to the first thickness and the second pitch is different from the first pitch, and/or iii) the second thickness is different from the first thickness and the second pitch is different from the first pitch.

[0129] Accordingly, the switchable reflective layer **220** without a voltage difference in the electrode layer **240** has a first reflectance and a first transmittance, and the switchable reflective layer **220** with a voltage difference in the electrode layer **240** has a second reflectance being different from the first reflectance and a second transmittance being different from the first transmittance.

[0130] In an aspect of the present disclosure, when the display panel **210** is not driven, i.e., an image non-display mode, there is no voltage difference in the electrode layer **240** so that the switchable reflective layer **220** can have high reflectance. When the display panel **210** is driven, i.e., an image display mode, there is a voltage difference in the electrode layer **240** so that the switchable reflective layer **220** can have high transmittance.

[0131] In an aspect of the present disclosure, when the display panel **210** is not driven, i.e., an image non-display mode, there is a voltage difference in the electrode layer **240** so that the switchable reflective layer **220** can have high reflectance. When the display panel **210** is driven, i.e., an image display mode, there is no voltage difference in the electrode layer **240** so that the switchable reflective layer **220** can have high transmittance.

[0132] Accordingly, the display device **200** of the present disclosure can provide excellent design without a transmittance decrease (i.e., a luminance decrease).

[0133] FIG. 8 is a schematic plane view showing an electroactive material pattern of a switchable reflective layer.

[0134] Referring to FIG. 8, the electroactive material pattern **260A** includes a plurality of first patterns **262** positioned at a corner of a first pattern region having a rectangular shape. Each of the plurality of first patterns **262** has a hexagonal shape with a major axis and a minor axis and is spaced apart from each other by a first distance (e.g., pitch) d_1 . The major axis length of the first

pattern **262** can be equal to or smaller than the first distance **d1**. In an aspect of the present disclosure, the major axis length of the first pattern **262** can be smaller than the first distance **d1**.

[0135] In an aspect of the present disclosure, in each of the plurality of first patterns **262**, the major axis length can be in the range of 220 to 240 nm, the minor axis length can be in the range of 120 to 140 nm, and the first distance **d1** can be in the range of 240 to 260 nm. In addition, each of the plurality of first patterns **262** can have a thickness equal to the first pitch **d1**.

[0136] FIGS. **9A** and **9B** are graphs showing reflection and transmittance of a visible ray in a switchable reflective layer. In FIG. **9A**, the x-axis (i.e., the horizontal axis) is the wavelength, and the y-axis (i.e., the vertical axis) is the reflectance. In FIG. **9B**, the x-axis (i.e., the horizontal axis) is the wavelength, and the y-axis (i.e., the vertical axis) is the transmittance.

[0137] As shown in FIG. **9A**, in the switchable reflective layer **220** including the electroactive material pattern **260A**, the switchable reflective layer **220** without a voltage to the electrode layer **240** has high reflectance to a visible light in the range of 500 to 550 nm. On the other hand, as shown in FIG. **9B**, the switchable reflective layer **220** with a voltage to the electrode layer **240** has high transmittance to the visible light.

[0138] For example, referring to FIG. **7A**, the electroactive material pattern **260A** without a voltage to the electrode layer **240** has a first thickness **t1**, a first width **w1** and a first pitch **p1** so that the switchable reflective layer **220** has high reflectance to a visible light in the range of 500 to 550 nm. On the other hand, the electroactive material pattern **260A** without a voltage to the electrode layer **240** has a second thickness **t1** smaller than the first thickness **t1** so that the reflectance of the switchable reflective layer **220** is decreased and the transmittance of the switchable reflective layer **220** is increased.

[0139] FIG. **10** is a schematic plane view showing an electroactive material pattern of a switchable reflective layer.

[0140] Referring to FIG. **10**, the electroactive material pattern **260B** includes a plurality of second patterns **264** positioned at a corner of a second pattern region having a triangular shape. Each of the plurality of second patterns **264** has a triangular shape and is spaced apart from each other by a second distance **d2**. A side length of the second pattern **264** can be equal to or smaller than the second distance **d2**. In an aspect of the present disclosure, the side length of the second pattern **264** can be equal to the second distance **d2**.

[0141] In an aspect of the present disclosure, in each of the plurality of second patterns **264**, each of the side length and the second distance **d2** can be in the range of 330 to 350 nm. In addition, each of the plurality of second patterns **264** can have a thickness smaller than the second distance **d2**. For example, each of the plurality of second patterns **264** can have a thickness of 190 to 210 nm.

[0142] FIGS. **11A** and **11B** are graphs showing reflection and transmittance of a visible ray in a switchable reflective layer. In FIG. **11A**, the x-axis (i.e., the horizontal axis) is the wavelength, and the y-axis (i.e., the vertical axis) is the reflectance. In FIG. **11B**, the x-axis (i.e., the horizontal axis) is the wavelength, and the y-axis (i.e., the vertical axis) is the transmittance.

[0143] As shown in FIG. **11A**, in the switchable reflective layer **220** including the electroactive material pattern **260B**, the switchable reflective layer **220** without a voltage to the electrode layer **240** has high reflectance to a visible light in the range of 630 to 700 nm.

[0144] On the other hand, as shown in FIG. **11B**, the switchable reflective layer **220** with a voltage to the electrode layer **240** has high transmittance to the visible light.

[0145] For example, referring to FIG. **7A**, the electroactive material pattern **260A** without a voltage to the electrode layer **240** has a first thickness **t1**, a first width **w1** and a first pitch **p1** so that the switchable reflective layer **220** has high reflectance to a visible light in the range of 500 to 550 nm. On the other hand, the electroactive material pattern **260A** without a voltage to the electrode layer **240** has a second thickness **t1** smaller than the first thickness **t1** so that the reflectance of the switchable reflective layer **220** is decreased and the transmittance of the switchable reflective layer **220** is increased.

[0146] In an aspect of the present disclosure, the switchable reflective layer **220** can include a first region and a second region, in each of a condition without a voltage to the electrode layer **240** and a condition with a voltage to the electrode layer **240**, the electroactive material pattern **260** in the first region and the electroactive material pattern **260** in the second region can have a difference in at least one of a shape, a thickness, a width and a pitch. As a result, in a condition without a voltage to the electrode layer **240** or a condition with a voltage to the electrode layer **240**, a light having a first wavelength range can be reflected by the switchable reflective layer **220** in the first region, and a light having a second wavelength range, which is different from the first wavelength range, can be reflected by the switchable reflective layer **220** in the second region.

[0147] For example, referring to FIGS. **8** and **10**, the first pattern region including the first patterns **262** can be arranged in the first region of the switchable reflective layer **220**, and the second pattern region including the second patterns **264** can be arranged in the second region of the switchable reflective layer **220**.

[0148] FIGS. **12A** and **12B** are graphs showing reflection and transmittance of a visible ray in a switchable reflective layer. In FIG. **12A**, the x-axis (i.e., the horizontal axis) is the wavelength, and the y-axis (i.e., the vertical axis) is the reflectance. In FIG. **12B**, the x-axis (i.e., the horizontal axis) is the wavelength, and the y-axis (i.e., the vertical axis) is the transmittance.

[0149] As shown in FIG. **12A**, the switchable reflective layer **220** without a voltage to the electrode layer **240** has high reflectance to a visible light in the range of 500 to 550 nm and the range of 630 to 730 nm. On the other hand, as shown in FIG. **12B**, the switchable reflective layer **220** with a voltage to the electrode layer **240** has high transmittance to the visible light.

[0150] As described above, the display device **200** of the present disclosure includes a display panel **210** and a switchable reflective layer **220** including an electrode layer **240** and an electroactive material pattern **260** and positioned at a side, i.e., a display side, of the display panel **210**. At least one of a thickness and a pitch of the electroactive material pattern **260** is changed according to a voltage to the electrode layer **240**. Accordingly, the switchable reflective layer **220** has a first reflectance and a first transmittance in a first mode and a second reflectance, which is smaller than the first reflectance, and a second transmittance, which is smaller than the first transmittance, in a second mode so that the display device **200** can have a desired design without the luminance decrease.

[0151] It will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the spirit or scope of the present disclosure. Thus, it is intended that the present disclosure cover the modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalents.

Claims

1. A display device, comprising: a display panel; and a switchable reflective layer including an electrode layer and an electroactive material pattern and positioned at a side of the display panel, wherein in a first mode in the electrode layer, the electroactive material pattern has a first thickness, a first width and a first pitch, wherein in a second mode in the electrode layer, the electroactive material pattern has a second thickness, a second width and a second pitch, and wherein the electroactive material pattern satisfies at least one of i) the second thickness being different from the first thickness, ii) the second width being different from the first width, and iii) the second pitch being different from the first pitch.
2. The display device according to claim 1, wherein in the first mode in the electrode layer, the switchable reflective layer has a first reflectance and a first transmittance, and wherein in the second mode in the electrode layer, the switchable reflective layer has a second reflectance being smaller than the first reflectance and a second transmittance being greater than the first transmittance.

3. The display device according to claim 2, wherein in the first mode in the electrode layer, the display panel is not driven, and wherein in the second mode in the electrode layer, the display panel is driven.
 4. The display device according to claim 2, wherein the second thickness is smaller than the first thickness.
 5. The display device according to claim 2, wherein the second pitch is smaller than the first pitch.
 6. The display device according to claim 1, wherein in the first mode in the electrode layer, the switchable reflective layer has a first reflectance and a first transmittance, and wherein in the second mode in the electrode layer, the switchable reflective layer has a second reflectance being greater than the first reflectance and a second transmittance being smaller than the first transmittance.
 7. The display device according to claim 6, wherein in the first mode in the electrode layer, the display panel is driven, and wherein in the second mode in the electrode layer, the display panel is not driven.
 8. The display device according to claim 6, wherein the second thickness is greater than the first thickness.
 9. The display device according to claim 6, wherein the second pitch is greater than the first pitch.
 10. The display device according to claim 1, wherein the electroactive material pattern includes at least one of poly vinylidene fluoride (PVDF), poly vinylidene cyanide (PVDCN), poly vinylidene chloride (PVDCL), polypyrrole, polyacrylonitrile, polydimethylsiloxane (PDMS), amide polymer, urea polymer and carbonyl polymer.
 11. The display device according to claim 1, wherein the electroactive material pattern includes a material represented by one of Formulas 1-1 to 1-3, 2, 3, 4-1, 4-2, 5-1, 5-2, 6-1 and 6-2:
##STR00002## wherein in each of Formulas 1-1, 1-2 and 2 to 6-2, n is an integer of 100 to 20000, wherein in Formula 1-3, a is a number of 0.7 to 0.8, b is a number of 0.2 to 0.3, and a summation of a and b is 1.
 12. The display device according to claim 1, wherein the switchable reflective layer includes a first region and a second region, and wherein the electroactive material pattern in the first region and the electroactive material pattern in the second region have a difference in at least one of a shape, a thickness, a width and a pitch.
 13. The display device according to claim 12, wherein a light having a first wavelength range is reflected by the switchable reflective layer in the first region, and a light having a second wavelength range being different from the first wavelength range is reflected by the switchable reflective layer in the second region.
 14. The display device according to claim 1, wherein the electrode layer includes a first electrode and a second electrode spaced apart from the first electrode.
 15. The display device according to claim 14, wherein each of the first and second electrodes includes a transparent conductive material.
 16. The display device according to claim 1, wherein the switchable reflective layer further includes an insulating layer covering the electrode layer and positioned under the electroactive material pattern.
 17. The display device according to claim 1, wherein the switchable reflective layer further includes a base covering the display panel and positioned under the electrode layer.
 18. The display device according to claim 1, wherein the display panel is an organic light emitting diode panel or a liquid crystal panel.
-